



It is imperative that an unbiased and accurate risk assessment, be carried out for any emerging technology, in order to get a holistic picture of its potential benefits and hazards. With the advent of new technologies, including nanotechnology, one should consider the potential unintended consequences to human health and the environment that might accompany development and use of the technology.

The National Nanotechnology Initiative (NNI) came out with the “Nanotechnology Environmental, Health, and Safety (EHS) Research Strategy” in 2011. The research integrated several important concepts – risk assessment and product life cycle stages – into the basic and applied research to understand the EHS impacts of nanomaterials.

Exposure to nanomaterials may occur unintentionally in the environment or through use of nanotechnology-enabled products. The figure above integrates the key risk assessment components – exposure assessment, hazard identification, and dose response (left-hand column) – with nanomaterial life cycle stages – from raw materials through commercialisation and end of product life (across the top) – and the exposure-effects pathways. The concentration of nanomaterials in the environment will depend on factors such as the nature and amount of the nanomaterial released, its physical and chemical properties, and time. The nanomaterials released into the environment may undergo transformation by various environmental conditions such as temperature and salinity, biological conditions such as habitat,